Practical Exam

Student_Score_Analysis

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=> Dataset
Student_ID, Age, Math_Score, Science_Score, English_Score, Hours_Studied, Pass_Fail (0/1)

PART-B

=> Libraries import
(i) numpy
(ii) pandas
(iii) scipy
(iv) matplotlib

=> Load the dataset
df = pd.read_csv ("student_scores.csv")
df.shape = (5000, 7)

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⇒ Dataset Shape

Rows :- df.shape[0]

Column :- df.shape[1]

Rows : 5000

Column : 7

df.info()

0	Student_ID	- Not null	- int	0.0
1	Age	- Not null	- int	0.0
2	Math_Score	- Not null	- int	0.0
3	Science_Score	- Not null	- int	0.0
4	English_score	- Not null	- int	0.0
5	Hours_Studied	- Not null	- float	0.0
6	Pass_Fail	- Not null	- in	0.0

df.describe()

Stat	Math	Science	English	Hours
count	5000	5000	5000	5000
mean	64.74	64.06	64.96	4.95
std	14.23	14.31	11.69	2.00
min	9.00	14.00	20.00	0.00
25%	55.00	55.00	57.00	3.60
50%	65.00	64.00	65.00	4.95
75%	74.00	74.00	73.00	6.30
max	100.00	100.00	100.00	12.00

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Step 1: Central Tendency & Dispersion

(a) Math-score → mean, median, mode

mean = $\Sigma x / n$
= 323702 / 5000
= 64.74

median : sort data asceding, n = 5000 (even)
median = (2500th + 2501st value) / 2
= (65 + 63) / 2
= 65

mode : value which comes most times
mode = 62 (appears 148 times)

(b) Science-Score → Range, Variance, SD

Range = max - min
= 100 - 14
= 86

Variance $s^2 = \frac{\Sigma (x - mean)^2}{(n-1)}$

mean = 64.06
$\Sigma (x - mean)^2 = 1022968.71$
$s^2 = 1022968.71 / 4999$
= 204.63

Standard Deviation $s = \sqrt{s^2}$

$= \sqrt{204.63}$

$= 14.31$

Interpretation:

Mathmeam, median, mode are close, so scores are almost symmetric. Science scores spread about 14 marks around the average.

Step-2: Probability basics

(a) Probability of passing

P(Pass) = Pass students / Total students

= 3440 / 5000

= 0.6880 = 68.80 %.

(b) Contingency table (Pass-Fail vs Hours-studied > 5)

Pass-Fail	Hours ≤ 5	Hours > 5	Total
Fail (0)	1367	193	1560
Pass (1)	1237	2203	3440
Total:	2604	2396	5000

(c) Conditional Probability

$P(\text{Pass} \mid \text{Hours} > 5) = \frac{P(\text{Pass} \cap \text{Hours} > 5)}{P(\text{Hours} > 5)}$

$= (2203/5000) / (2396/5000)$

$= 2203/2396$

$= 0.9194 \approx 91.94\ \%$

Step-3: Distribution & Visualization

(a) Histogram & Normal curve (math-score)

$n = 5000$, $\mu = 64.74$, $\sigma = 14.23$

Normal curve:

$f(x) = 1/(\sigma\sqrt{(2\pi)}) \times e^{(-(x-\mu)^2/2\sigma^2)}$

At $= \mu$: $f(\mu) = 1/(14.23 \times 2.5066)$

$= 0.0280$

(b) Skewness & Kurtosis (Science-Score)

$n = 5000$, mean $= 64.06$, $\sigma = 14.30$

Skewness $= [\Sigma (x - \text{mean})^3 / n] / \sigma^3$

$= -67.45 / (14.30)^3$

$= -0.0230$

Kurtosis (excess) $= [\Sigma (x - \text{mean})^4 / n] / \sigma^4 - 3$

$= 122176.18 / (14.30)^4 - 3$

$= -0.0812$

(c) Q-Q Plot (English-score)

$n = 5000$, mean $= 64.96$, Std $= 11.69$

Skewness $= -0.0215$, Kurtosis $= -0.1318$

Steps: sort data → find theoritical normal quantiles → plot stats.probplot(x, plot=plt)

Observation: Points lie near the straight line, so English score are approximately normal. Small gap at the ends because marks are limited between 0 and 100.

Step-4: Linear Algebra mini Task

First 5 students as vectors:

Math vector $a = [75, 66, 72, 72, 56]$

Science vector $b = [54, 57, 58, 65, 63]$

(a) Dot product: $a \cdot b = \Sigma (a \times b)$

$= 75 \times 54 + 66 \times 57 + 72 \times 58 + 72 \times 65 + 56 \times 63$

$= 4050 + 3762 + 4176 + 4680 + 3528$

$= 20196$

(b) Norm of math vector

Norm 1: $\|a\|_1 = \Sigma |a|$

$= 75 + 66 + 72 + 72 + 56$

$= 341$

Norm 2: $\|a\|_2 = \sqrt{(\Sigma a^2)}$

$= \sqrt{(5625 + 4356 + 5184 + 3136)}$

$= \sqrt{23485}$

$= 153.25$

(c) Angle between two vectors

$\cos\theta = (a.b) / (\|a\| \times \|b\|)$

$\|b\| = \sqrt{(2916 + 3249 + 3364 + 4225 + 3969)}$

$= \sqrt{17723} = 133.13$

$\cos\theta = 20196 / (153.25 \times 133.13)$

$= 0.9899$

$\theta = \cos^{-1}(0.9899)$

$= 0.1421$ radian

$= 8.14^\circ$